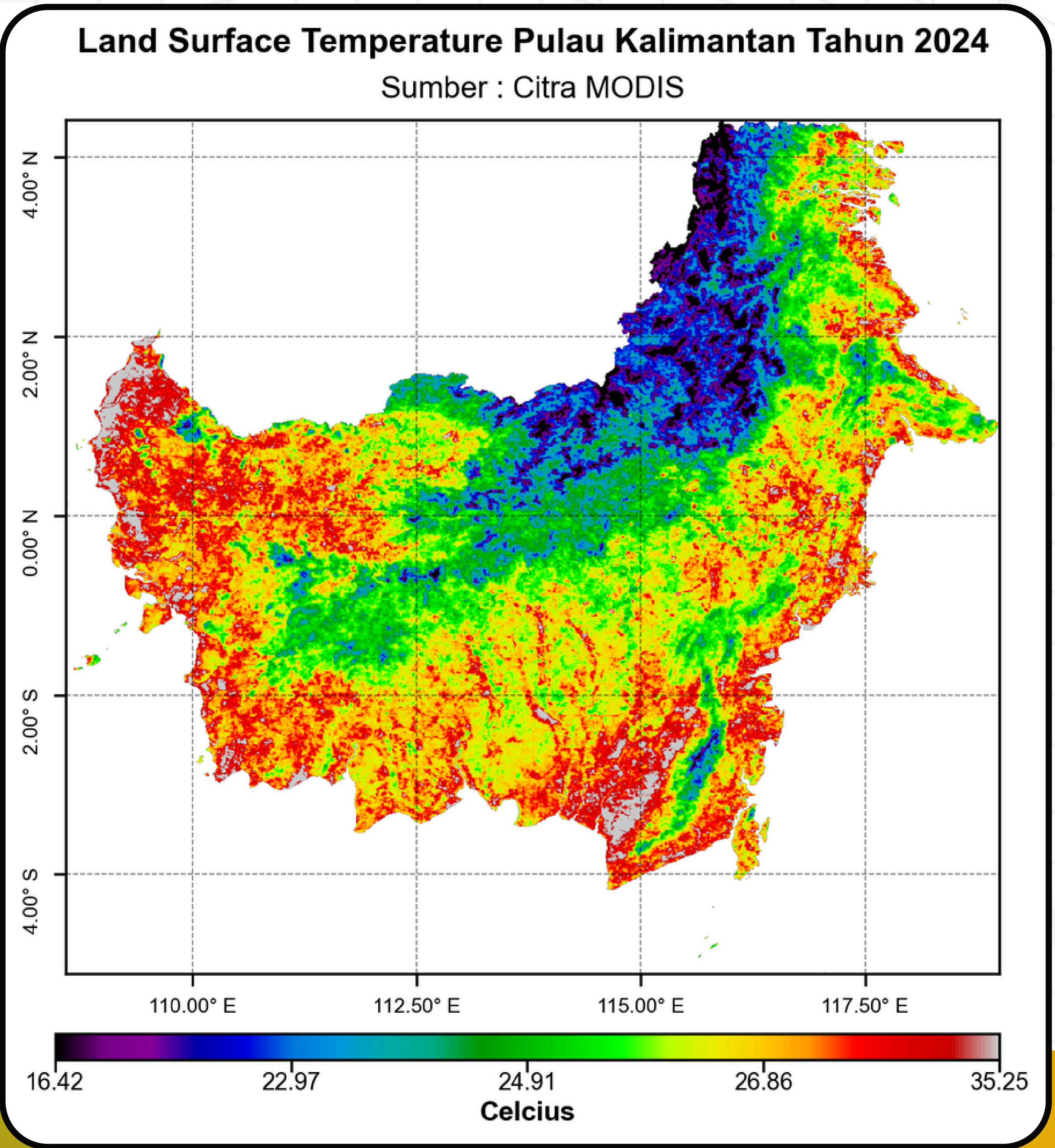
Publication-Ready Raster Visualization in QGISAchieving Matplotlib and rasterio Aesthetics Through a GUI

Step by step using RasterViz Plugin



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RasterViz Plugin

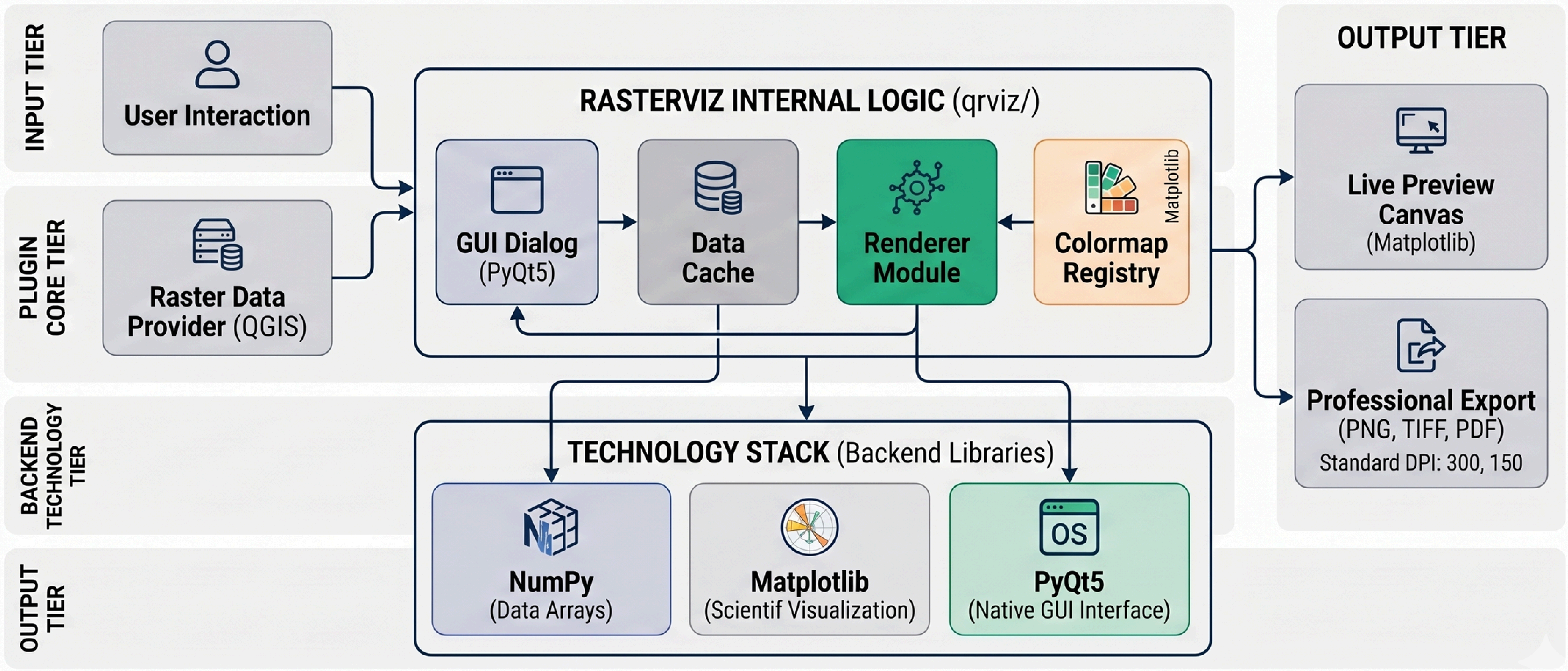
RasterViz is a QGIS plugin that helps produce raster figures styles commonly found in matplotlib and rasterio workflows.

The idea behind this plugin actually started from something simple: I really like the style of raster visualizations often seen in remote sensing papers, especially stretchbars with fully readable labels and cleaner scientific layouts instead of only showing minimum and maximum values. I always found those visualization styles more informative and aesthetically pleasing for scientific communication.

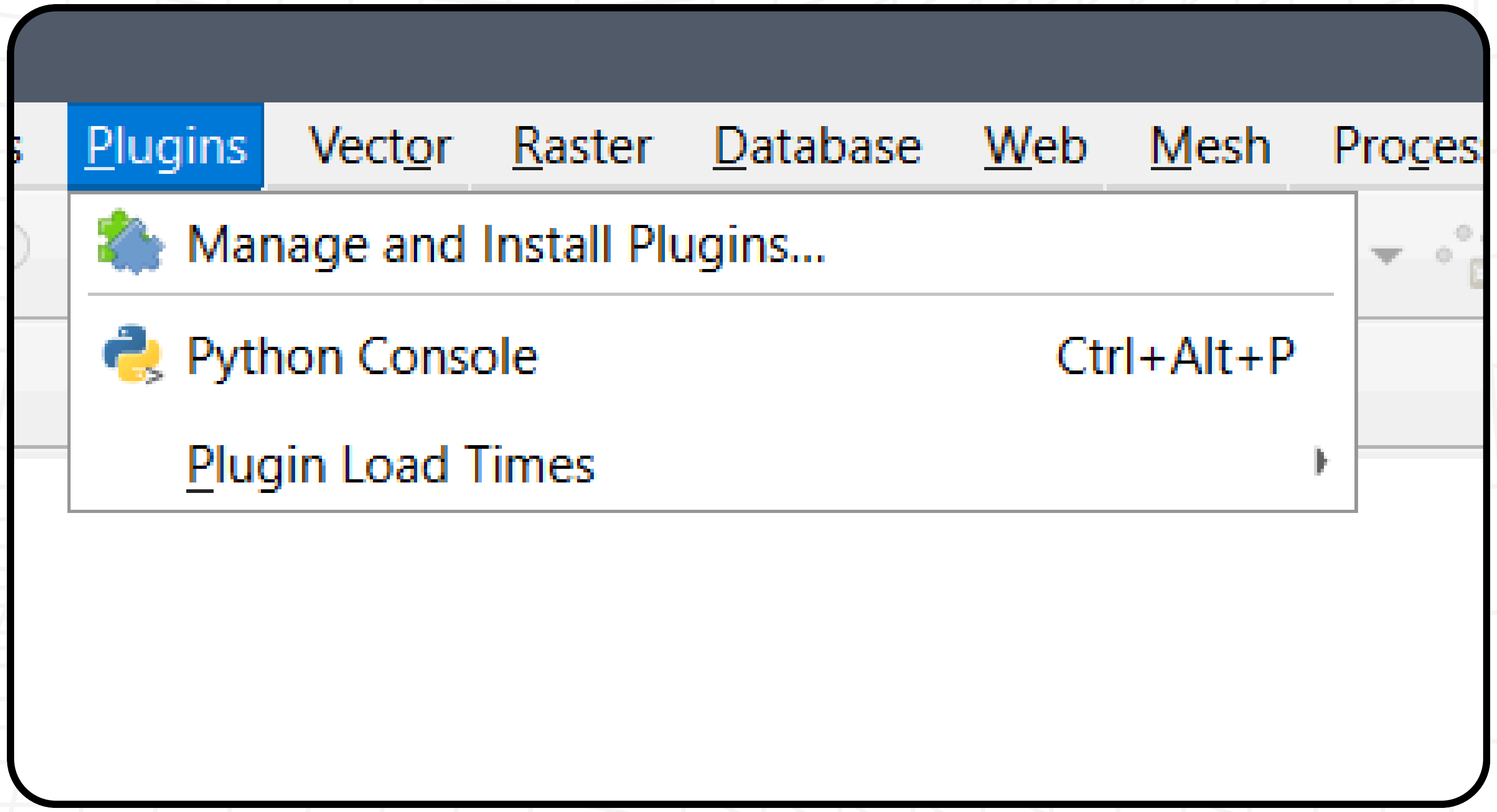
https://plugins.qgis.org/plugins/qrviz/

https://github.com/Defani/QRasterVIZ

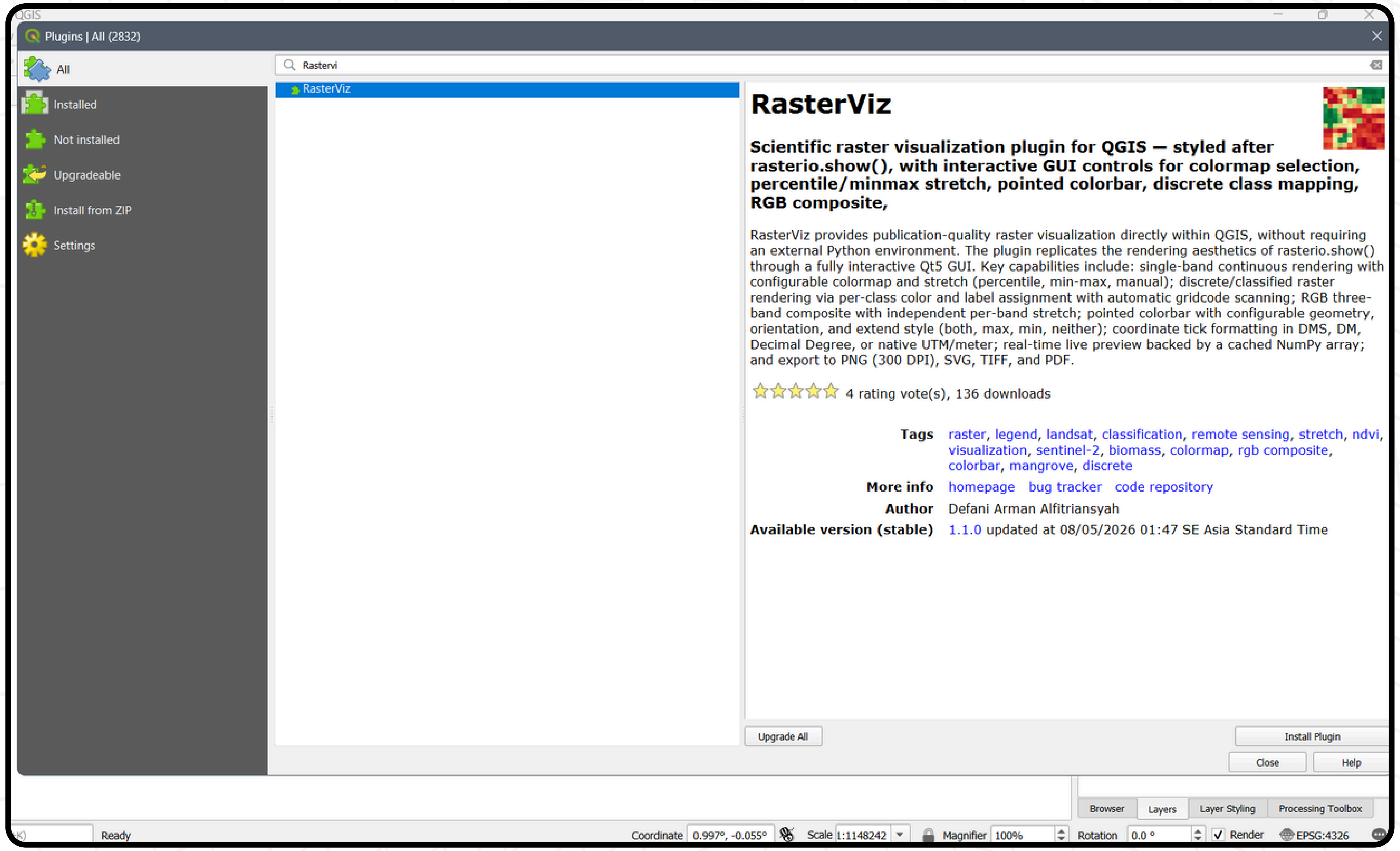
Architecture



Instalation

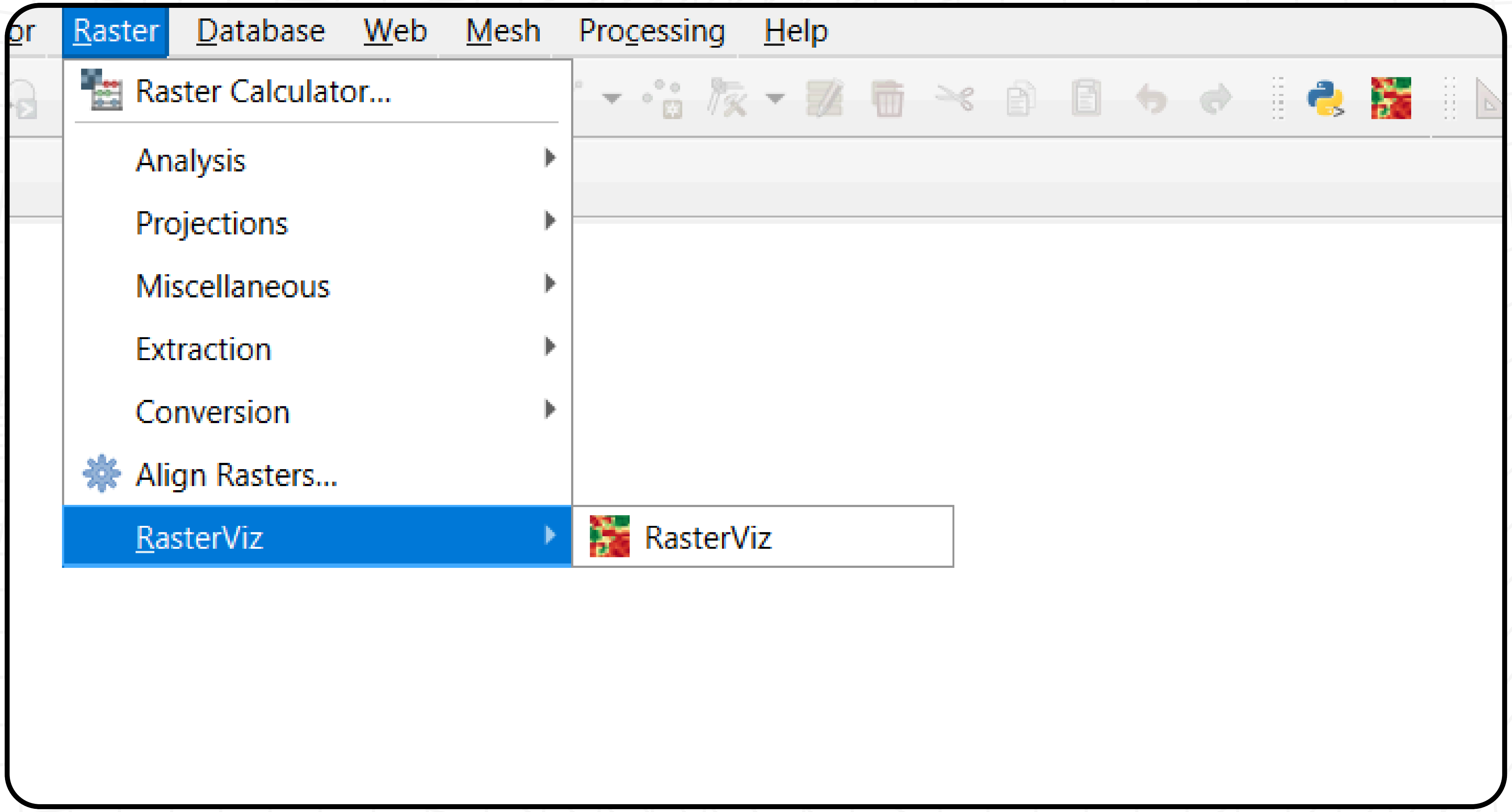


Open QGIS then in the plugin menu, click manage and install plugin

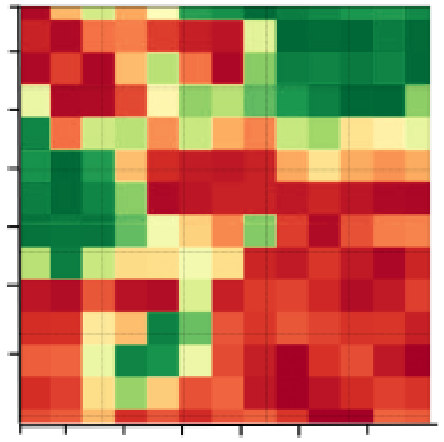


In the plugin panel, select "All" then search for "RasterViz" and press "install", wait until the installation is complete then close the plugin manager panel.

Run The Plugin



After the installation is complete, to run the plugin, you can press the menu in the form of the plugin logo, or it can be accessed via the raster toolbar menu.



RasterVis icon

Interface

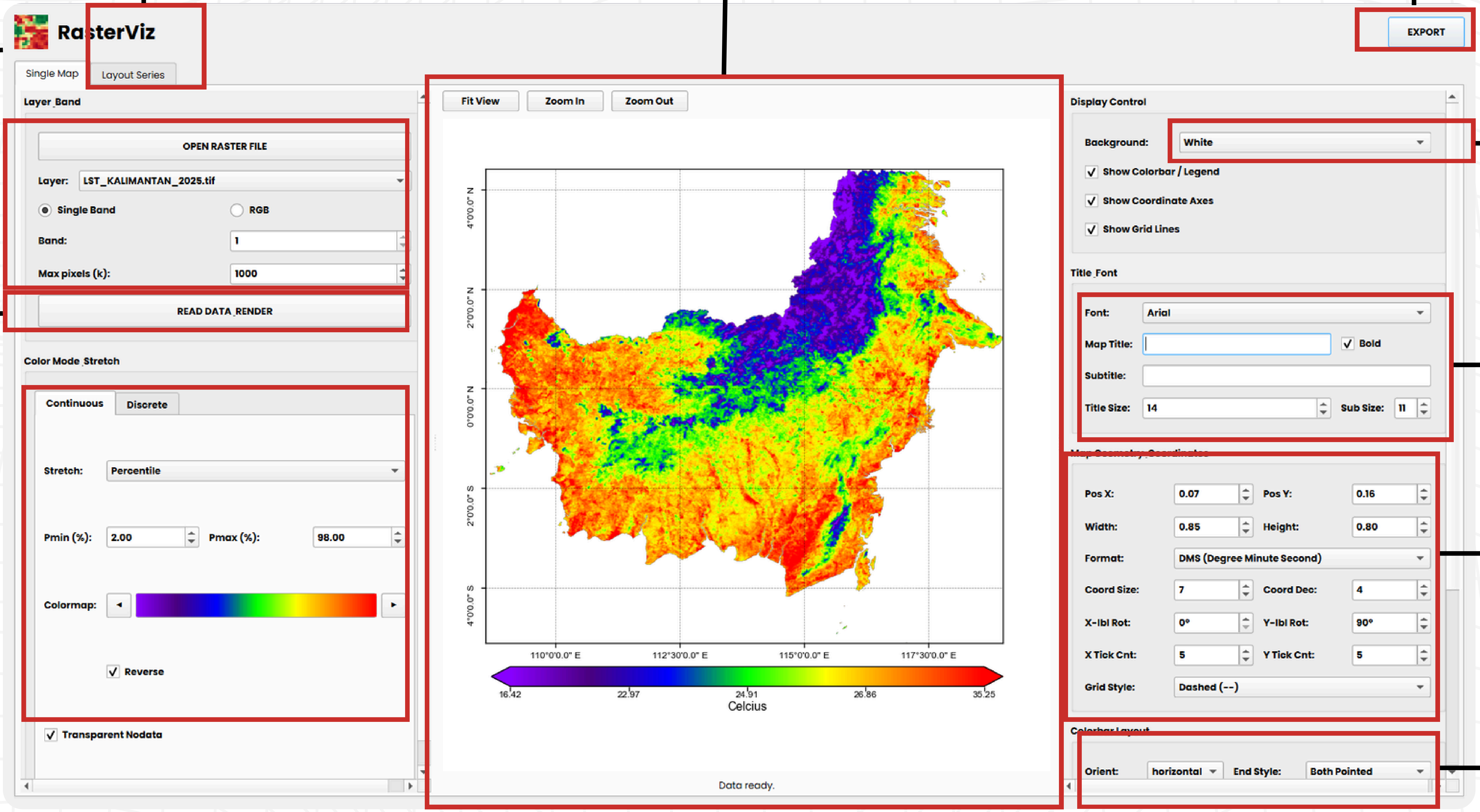
Tab menu for multiple image

Map Preview

Export

1. Import File
2. Raster type load mode (continous, imagery composite, DIscrete)

Preview activate



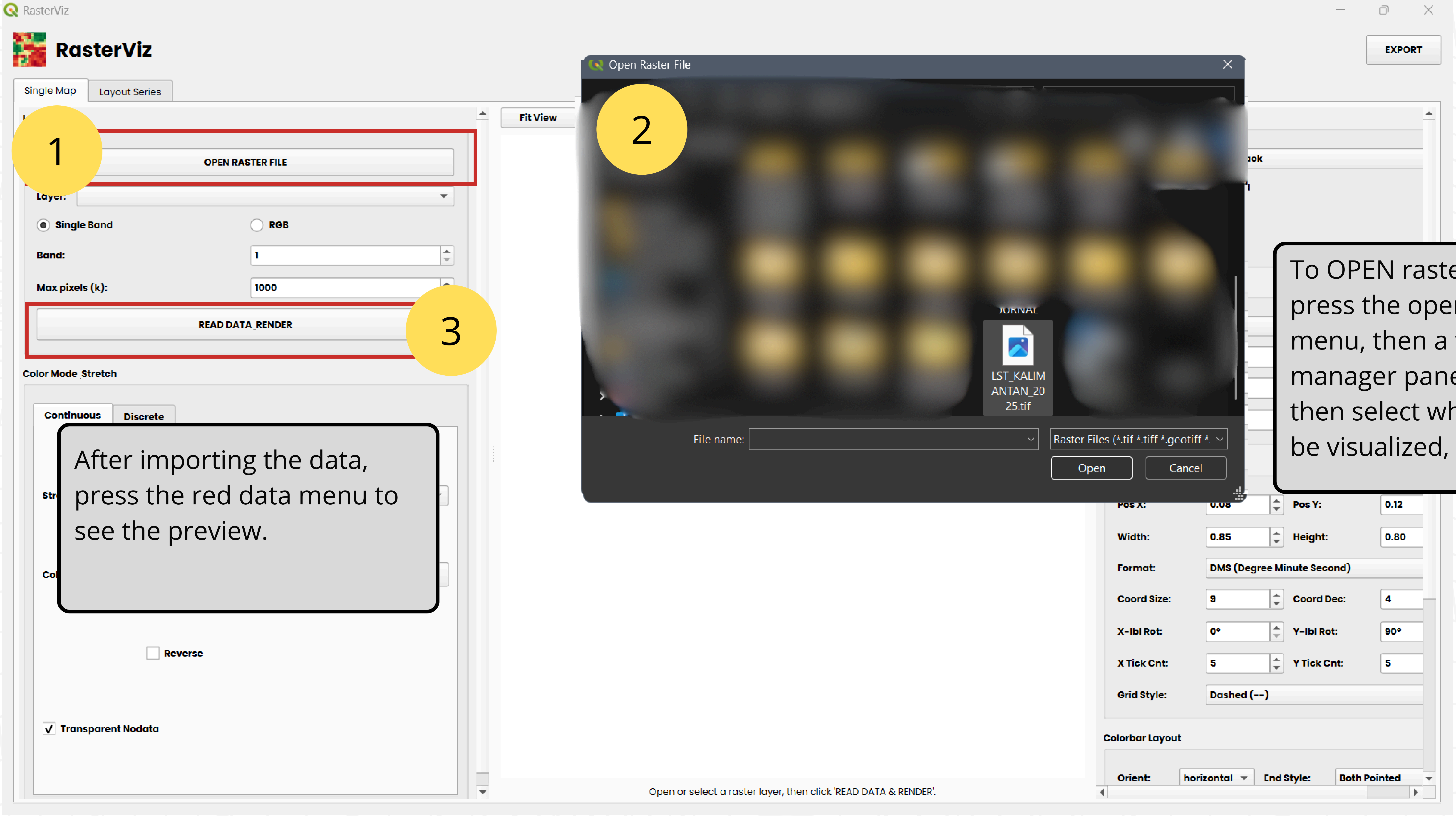
Canvas Color settings

Main title settings

coordinat label settings

streech Bar Settings

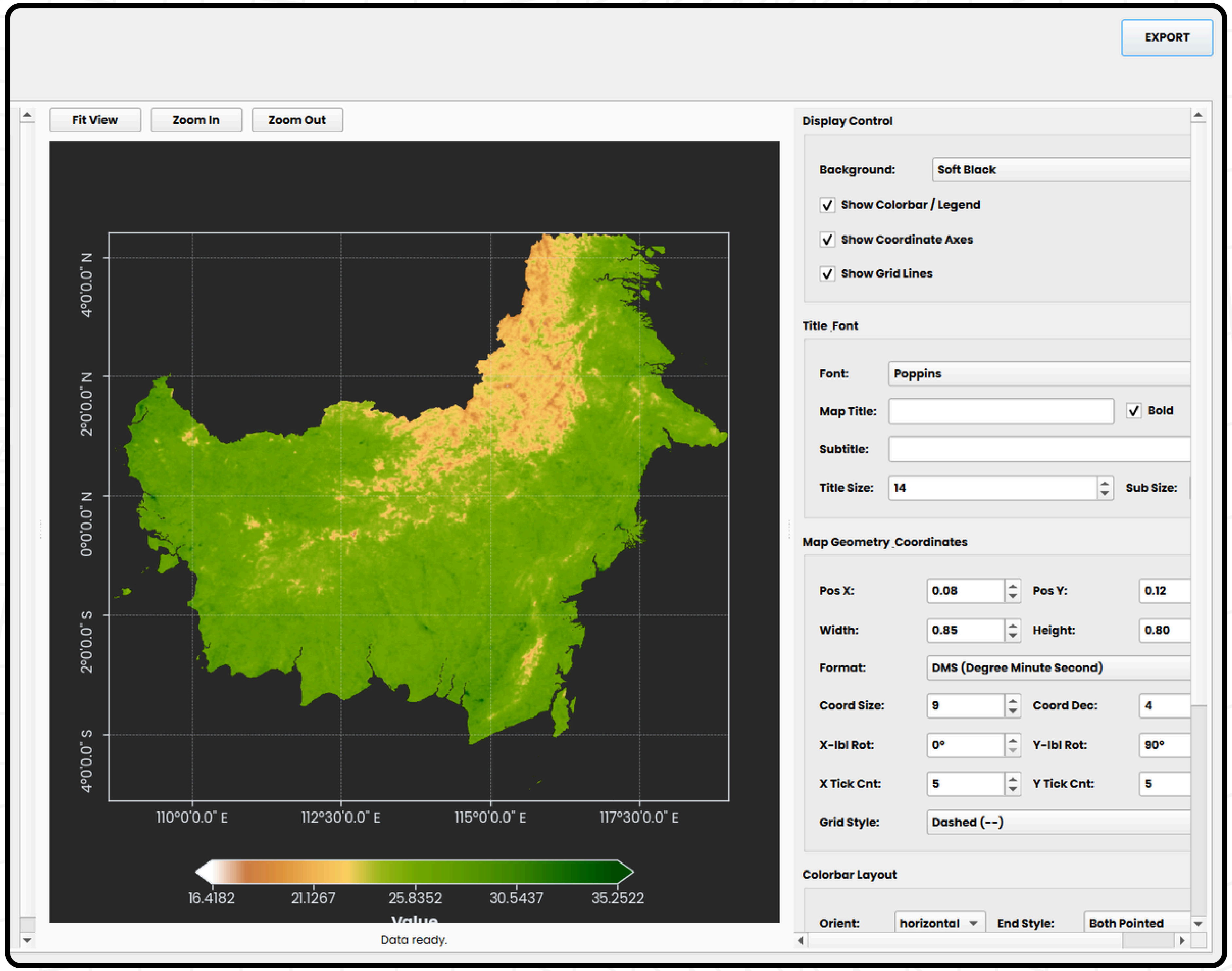
Import and Read Contionous Raster Data

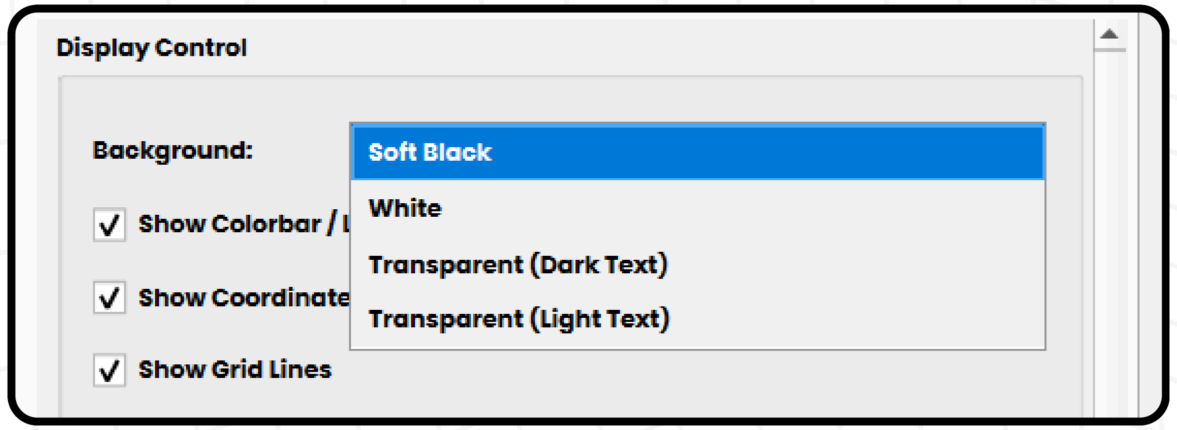


After importing the data, press the red data menu to see the preview.

To OPEN raster data, please press the open raster file menu, then a floating file manager panel will appear, then select which data will be visualized, and press OK.

Sets the canvas color

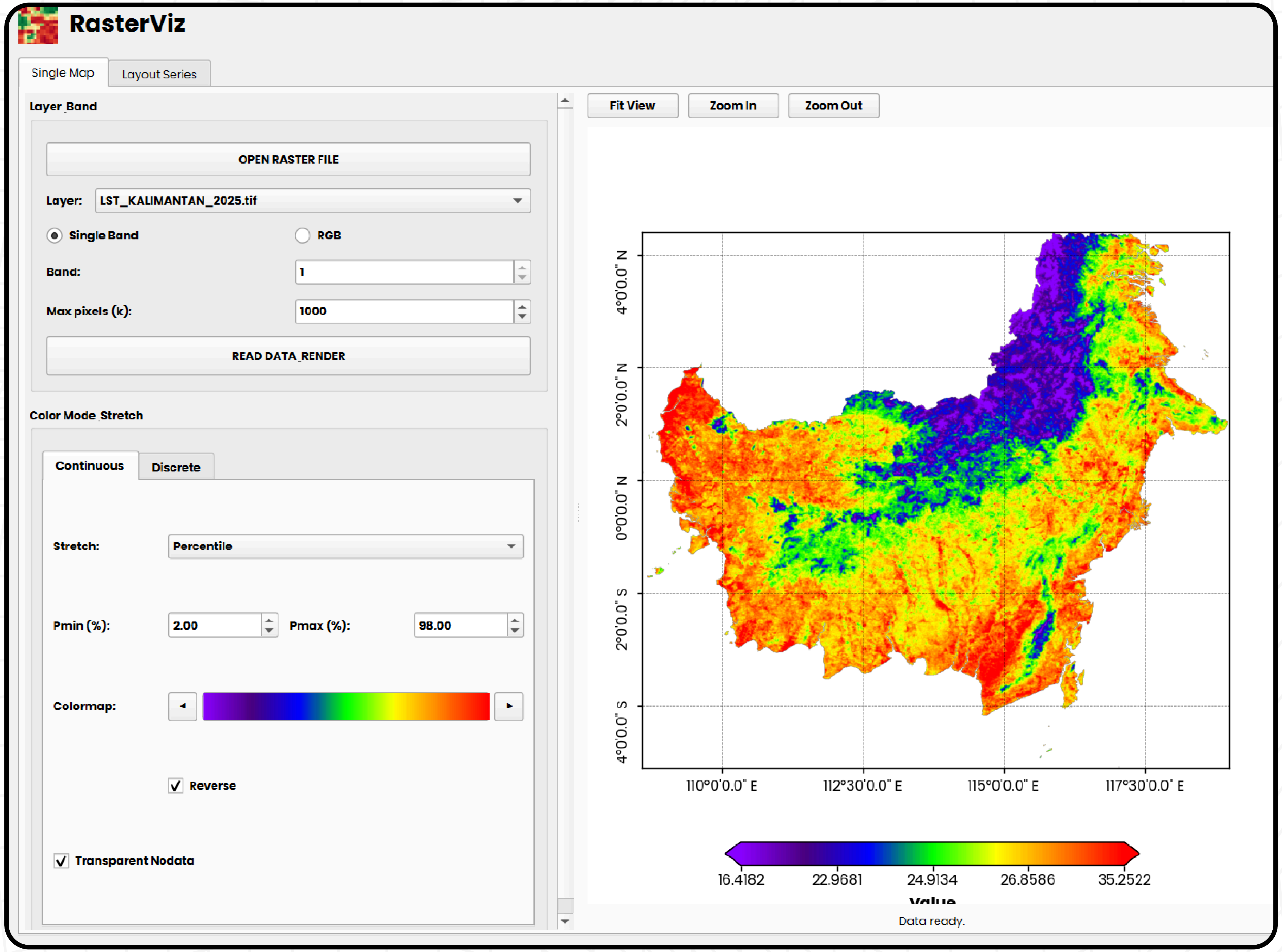


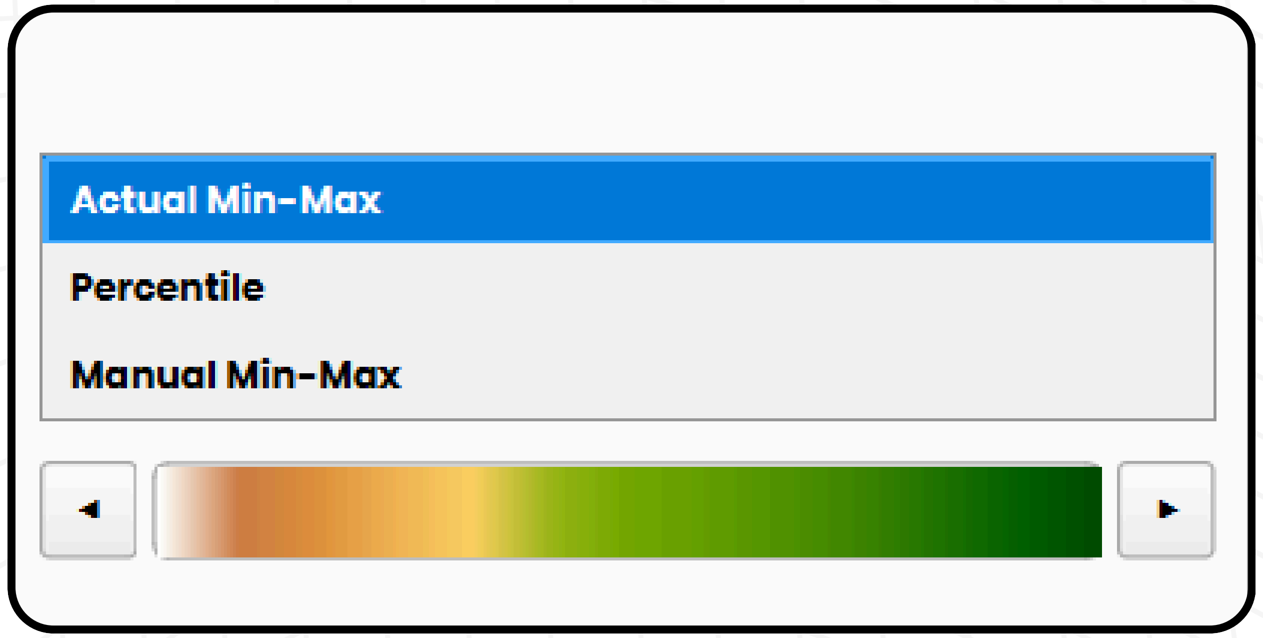


By default, when you click "Read Data," the data will be presented in a complete visualization (layout, color, and stretch bar legend).

If you want to change the background, you can click the "Background" menu and select the color you want to use.

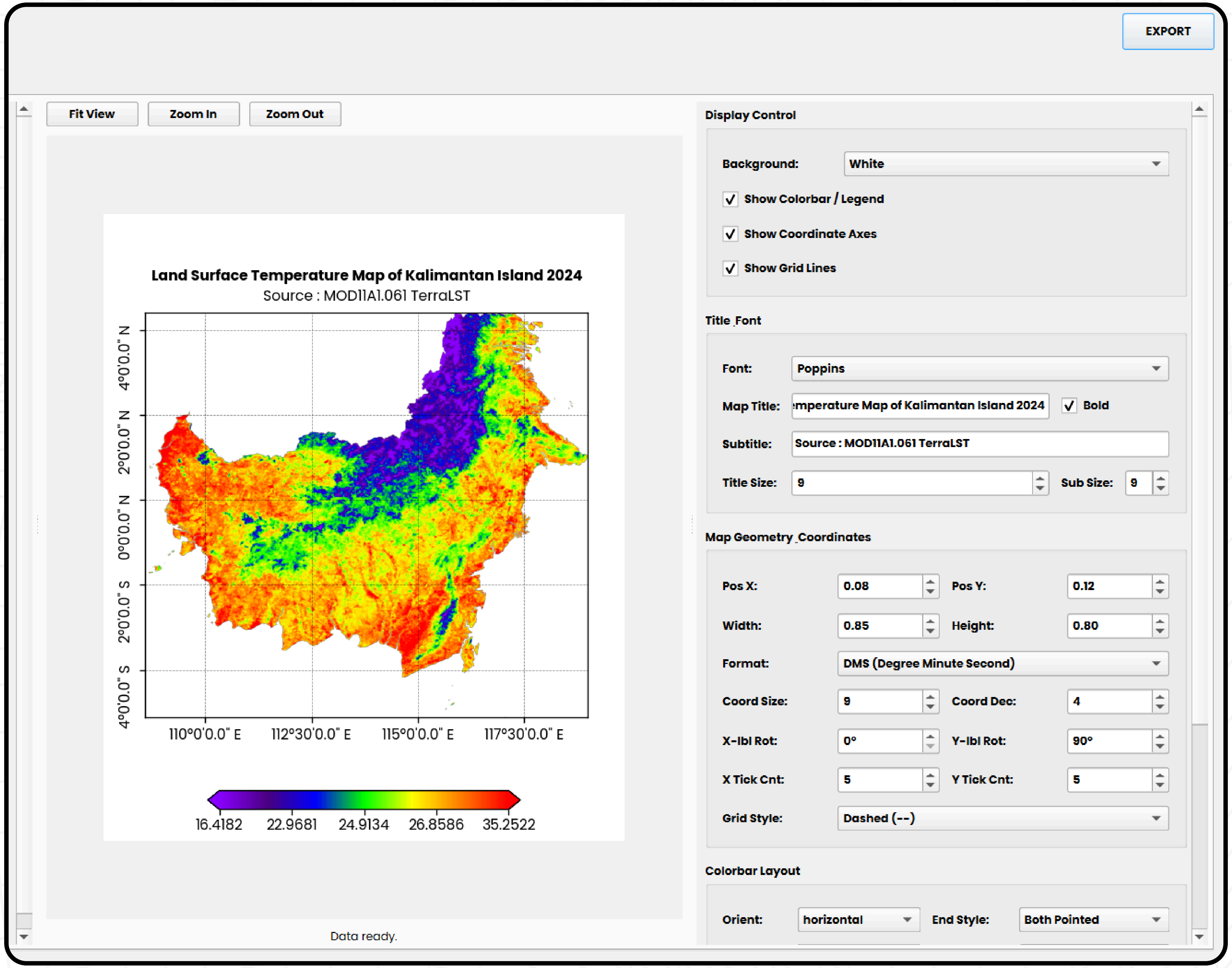
Raster Color





Use ◀ ▶ to switch color, you also can reverse the color

Maps Title



Font: Poppins

Font family applied to the map title, subtitle, and all coordinate tick labels. Lists all fonts installed on your system.

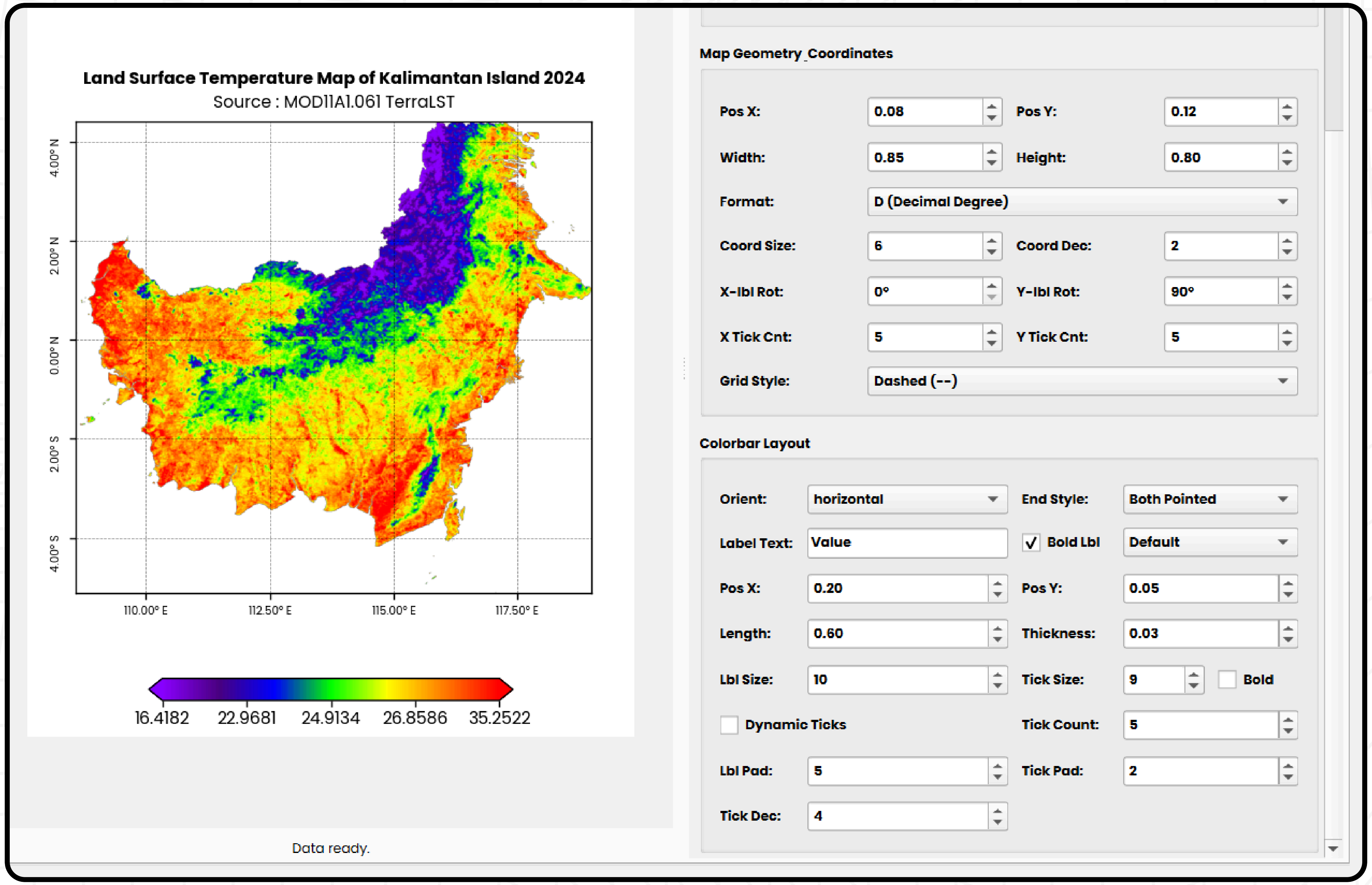
Map Title: ımperature Map of Kalimantan Island 2024 ✓ **Bold**

Primary title text displayed at the top of the figure. The title is draggable in drag mode — click and drag it anywhere on the canvas. Default: empty (no title rendered)

Subtitle: Source : MOD11A1.061 TerraLST

Secondary text line below the title. Typically used for date, sensor, or study area information (e.g. "Sentinel-2, Band 8, 15 March 2024"). Also draggable independently from the title.

Map Geometry Coordinates



Grid Style

Line style for grid overlays. Solid ; Dashed s; Dotted

All position values are fractions of the figure (0 to 1). The standard setup is: Pos X=0.08, Pos Y=0.12, Width=0.85, Height=0.80 — this leaves space at the bottom (Y=0.05 to 0.10) for a horizontal colorbar. The X-label Rotation at 45° prevents overlap when longitude ticks are dense. Setting Y-label Rotation to 0° makes latitude labels horizontal and easier to read. Coord Decimals controls how many decimal places appear in tick labels.

Pos X

Left edge of the map axes as a fraction of the figure width (0 = far left, 1 = far right).
Default: 0.08 | Range: 0 – 1

Pos X

Bottom edge of the map axes as a fraction of the figure height (0 = bottom, 1 = top). Increase to make room for a horizontal colorbar below.
Default: 0.12 | Range: 0 – 1

Format

Coordinate label format. DMS: 106°49'0" E. DM: 106°49.0' E. D: 106.8167° E. Default (UTM): raw numeric value in metres or degrees.

Coord Size

Font size of coordinate tick labels in points.
Default: 9 pt | Range: 4 – 30 pt

Coord Dec

Decimal places shown in coordinate tick labels. Applied to Decimal Degree and UTM/Metre modes.
Default: 4 | Range: 0 – 8

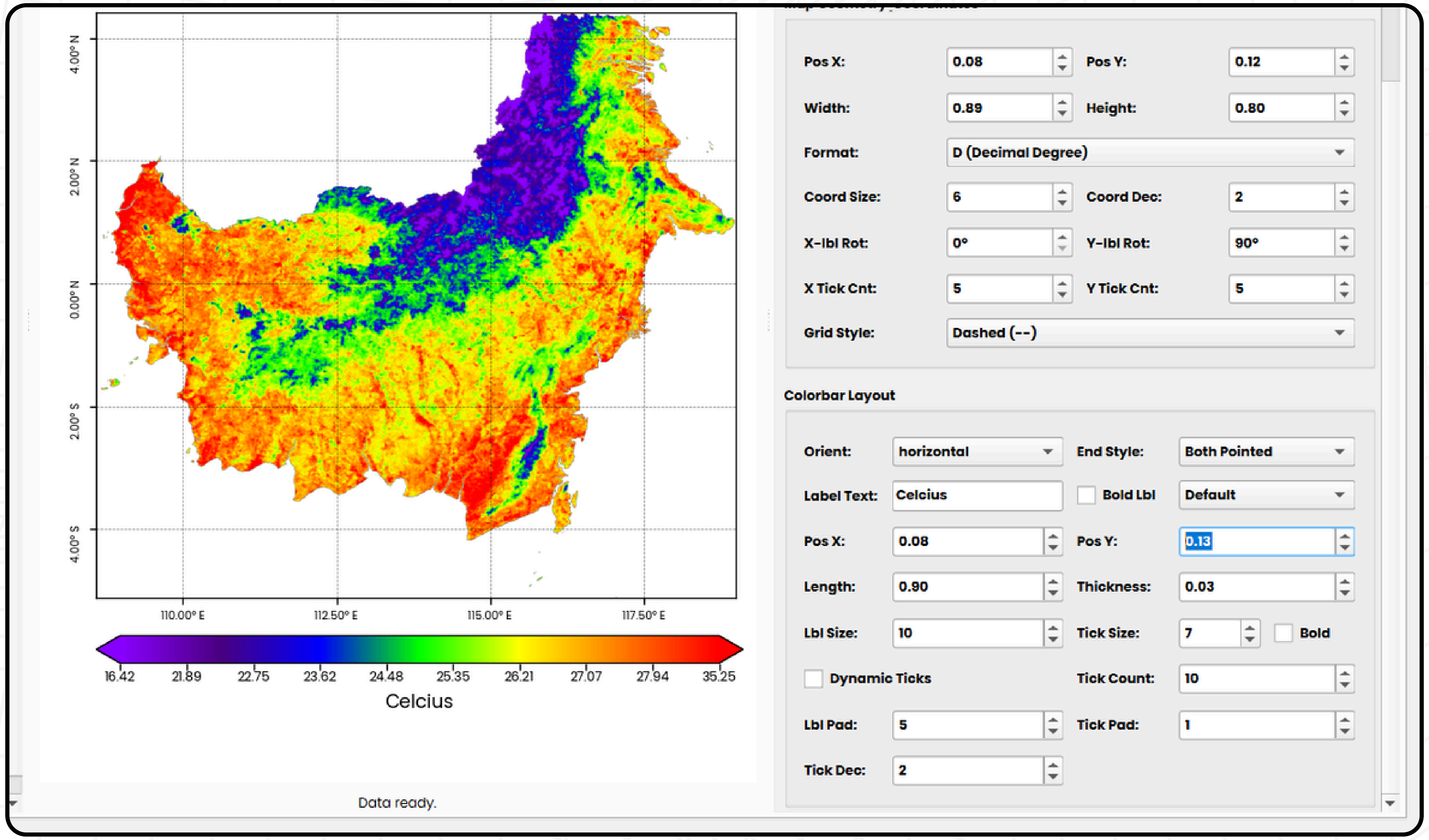
X-lbl Rotation

Rotation of X-axis (longitude) tick labels in degrees. Set to 45° to prevent label overlap when tick density is high.
Default: 0° | Range: 0 – 360°

X Tick Cnt

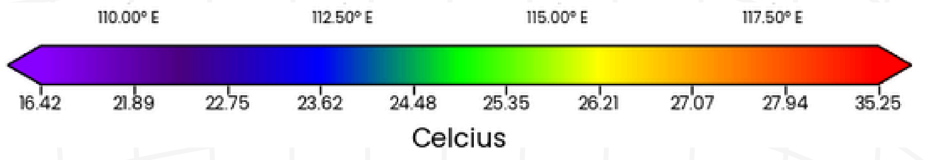
Maximum number of Y-axis tick intervals.
Default: 5 | Range: 2 – 20

Colorbar style

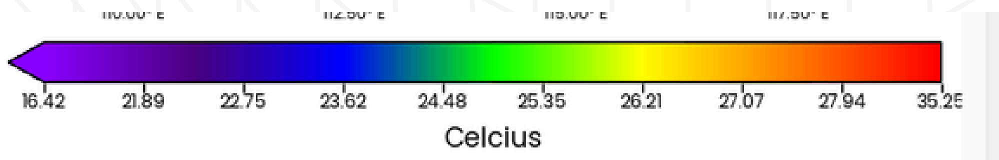


Style

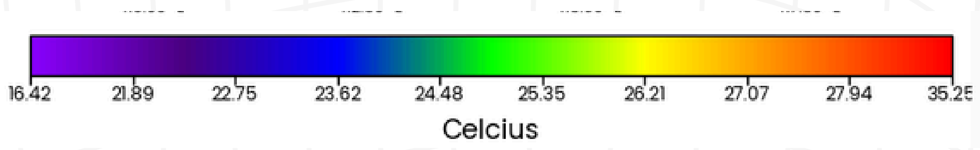
Both pointed

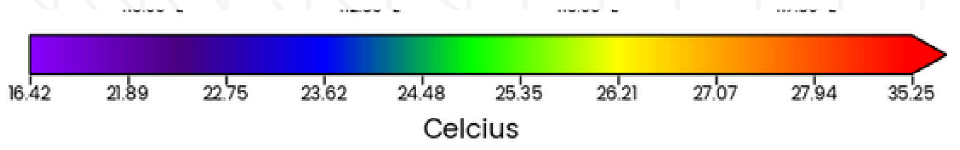


left pointed (min)



box standard





horizontal — bar runs left-to-right below the map. vertical — bar runs top-to-bottom at the right. Switching orientation auto-snaps position and length/thickness to sensible defaults.

Label

Text label for the colorbar axis, e.g. "NDVI", "Backscatter (dB)", "Carbon Stock (Mg/ha)", "Celcius".

Default: "Value"

Bold lbl

Applies bold weight to the colorbar label text independently from tick labels.

Default: checked

Pos X/Y ->x = left /right control , Y ->up/down

Left and bottom edges of the colorbar axes as fractions of the figure.

Default: X=0.20, Y=0.05

length

Length of the colorbar along its long axis as a fraction. For horizontal = width; for vertical = height. Increasing from 0.60 to 0.80 makes the bar span more of the figure width.

Default: 0.60 | Range: 0.01 – 1

Thickness

Thickness of the colorbar along its short axis. For horizontal = height; for vertical = width. Keep thin (0.02–0.05) for a clean scientific aesthetic.

Default: 0.03 | Range: 0.01 – 1

Lbl Size

Font size of the colorbar axis label (e.g. "NDVI").

Default: 10 pt | Range: 6 – 30 pt

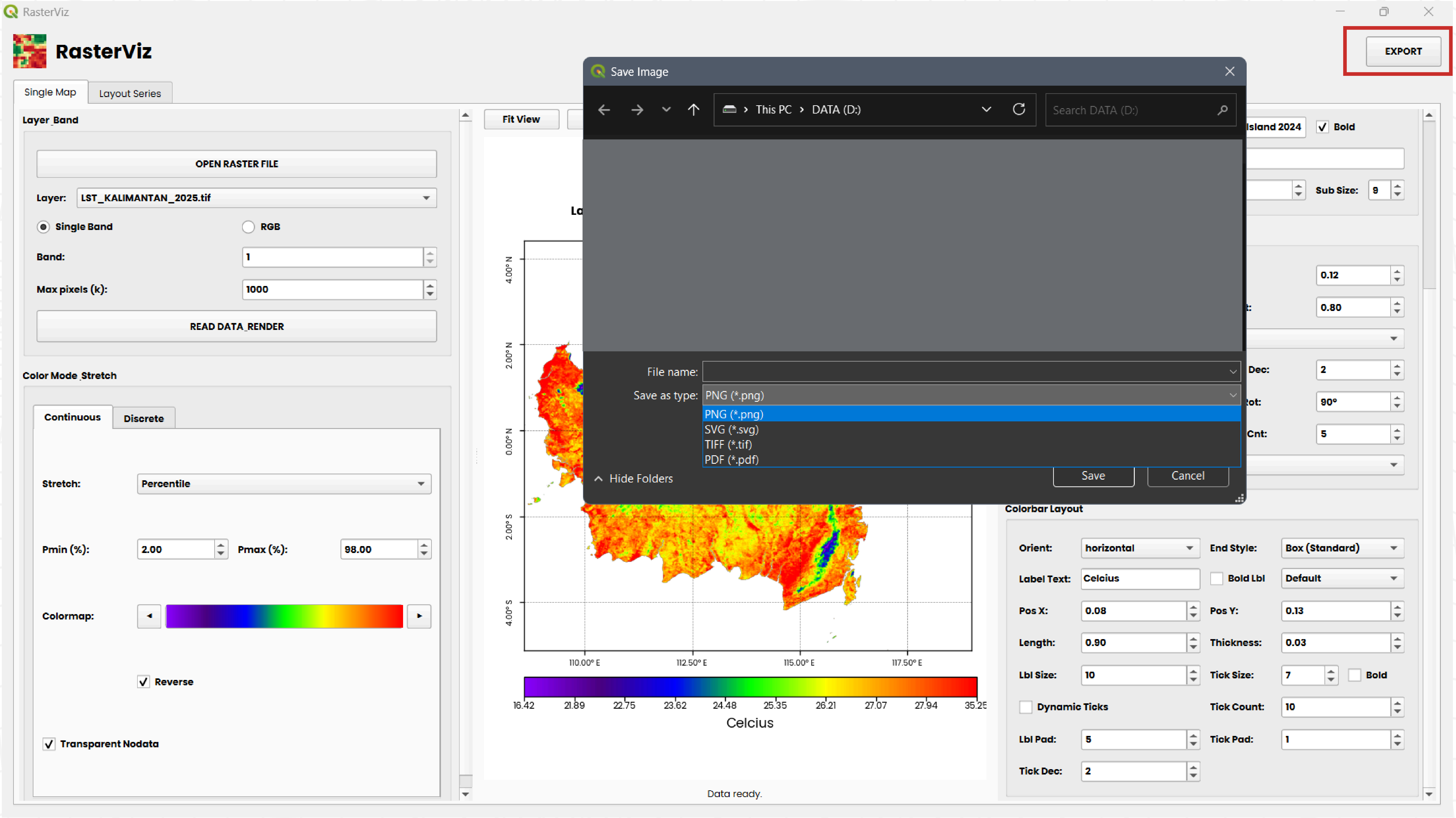
tic Size

Font size the colorbar tick labels (the numeric values).

Tick Count

Number of tick positions on the colorbar (including the two ends). 5 ticks = min, 25%, 50%, 75%, max — sufficient for reader interpolation.

Export



RasterViz
EXPORT
Single Map
Layout Series
Layer_Band
OPEN RASTER FILE
Layer: LST_KALIMANTAN_2025.tif
Single Band
RGB
Band: 1
Max pixels (k): 1000
READ DATA_RENDER
Color Mode_Stretch
Continuous
Discrete
Stretch: Percentile
Pmin (%): 2.00
Pmax (%): 98.00
Colormap:
Reverse
Transparent Nodata
Fit View
Save Image
This PC › DATA (D:)
Search DATA (D:)
File name:
Save as type: PNG (*.png)
PNG (*.png)
SVG (*.svg)
TIFF (*.tif)
PDF (*.pdf)
Hide Folders
Save
Cancel
Celcius
Data ready.
Colorbar Layout
Orient: horizontal
End Style: Box (Standard)
Label Text: Celcius
Bold Lbl
Default
Pos X: 0.08
Pos Y: 0.13
Length: 0.90
Thickness: 0.03
Lbl Size: 10
Tick Size: 7
Bold
Dynamic Ticks
Tick Count: 10
Lbl Pad: 5
Tick Pad: 1
Tick Dec: 2

Thank You

https://plugins.qgis.org/plugins/qrviz/

https://github.com/Defani/QRasterVIZ

https://medium.com/@defaniarman

https://github.com/Defani

https://rpubs.com/defanii

https://www.instagram.com/de.fanii